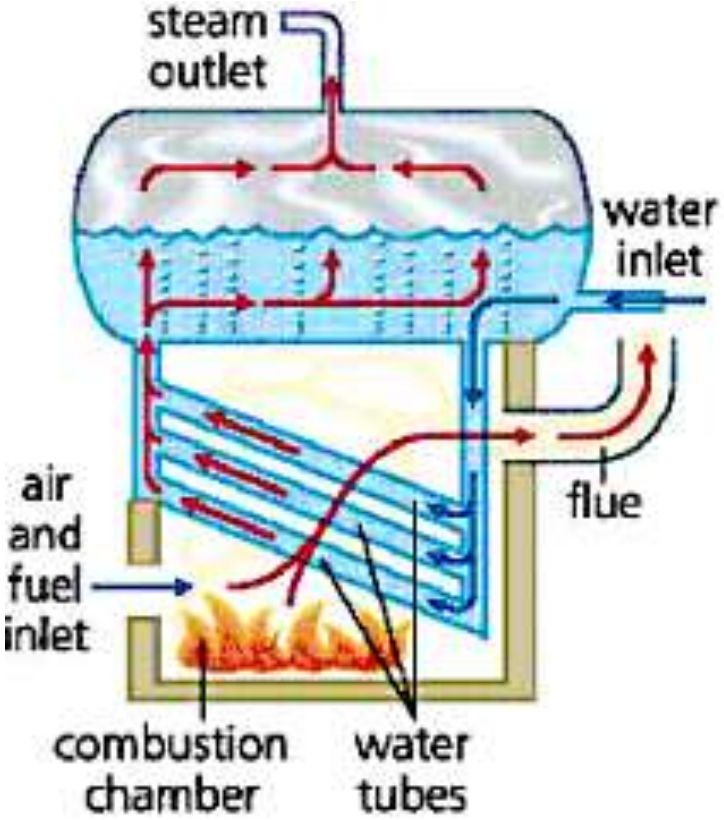
Boiler



Pressure Cooker





TYPES OF BOILER

There are mainly two types of boiler : water tube boiler and fire tube boiler.

- In fire tube boiler, there are numbers of tubes through which hot gases are passed and water surrounds these tubes.
- Water tube boiler is reverse of the fire tube boiler. In water tube boiler the water is heated inside tubes and hot gasses surround these tubes.

Fire tube boiler

Fire Tube Boiler is the type of boiler in which the water is flowing outside (Surrounded) the tube and hot gases which are produced by the combustion chamber are inside the tube and the further boiler is used to rotate the turbine blade and that used for the generation of electricity.

Cochran Boiler:

Cochran Boiler is a modification of a simple vertical boiler and a Portable type boiler. Coal or Oil is used here to produce the hot flue gasses. The steam generated from the Cochran boiler is used in Hydro Power Plant and some small industries too.

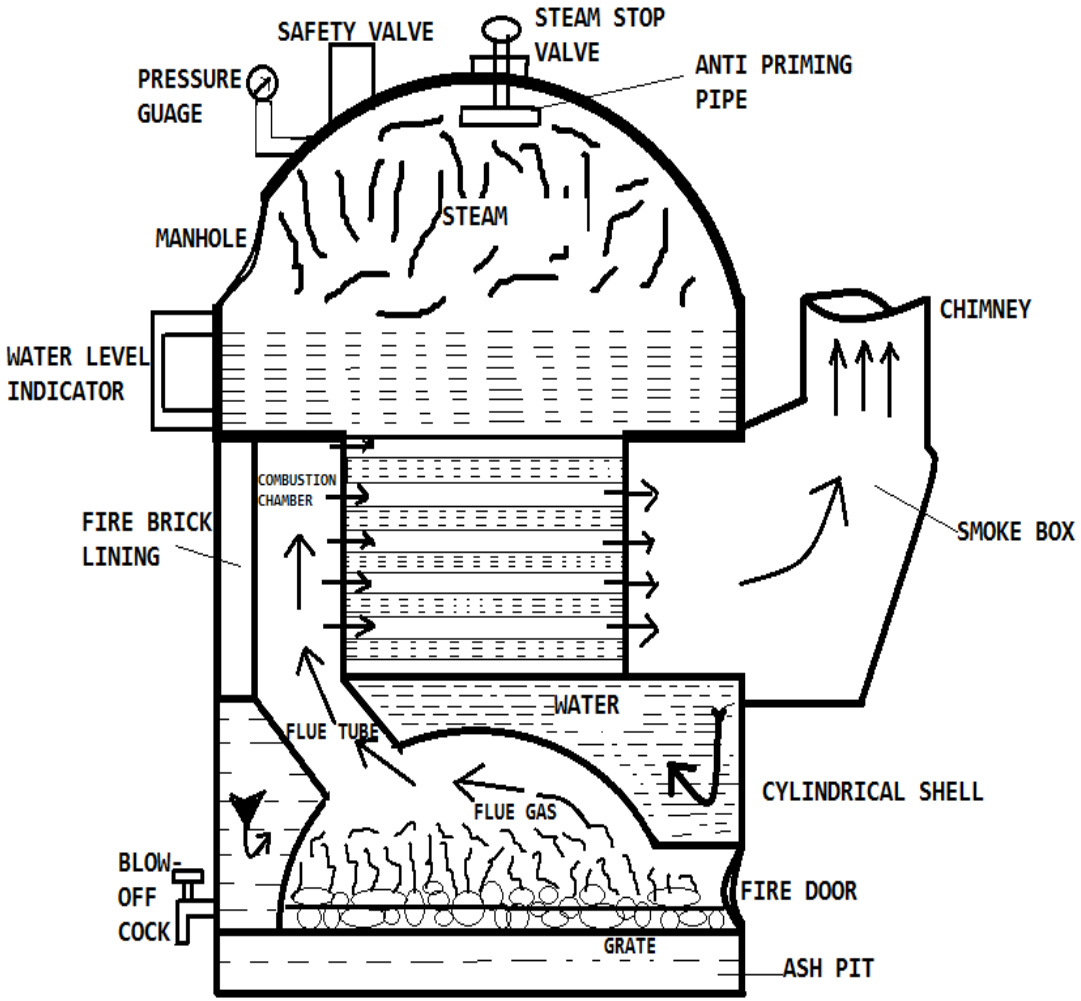


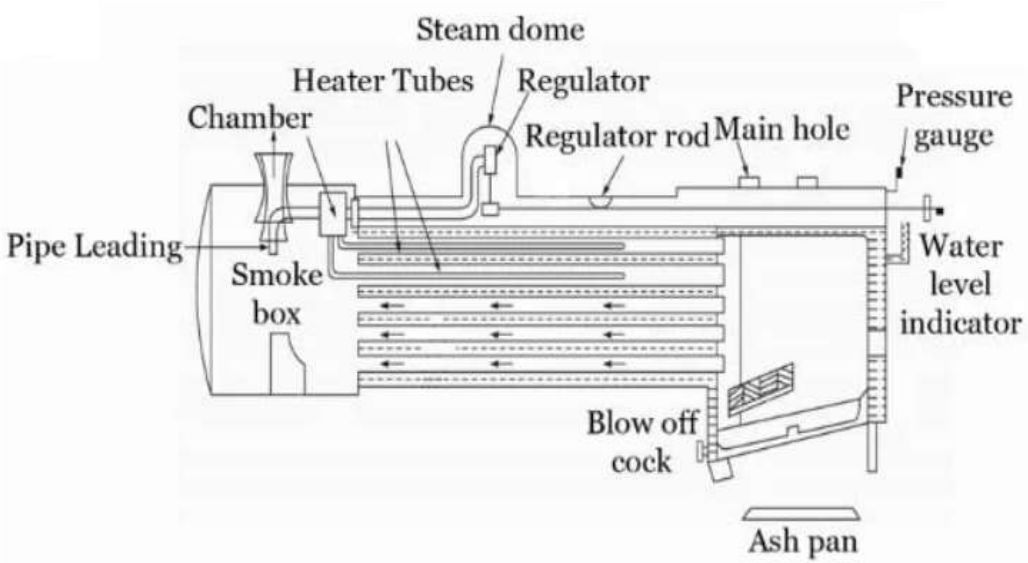
FIG: COCHRAN BOILER

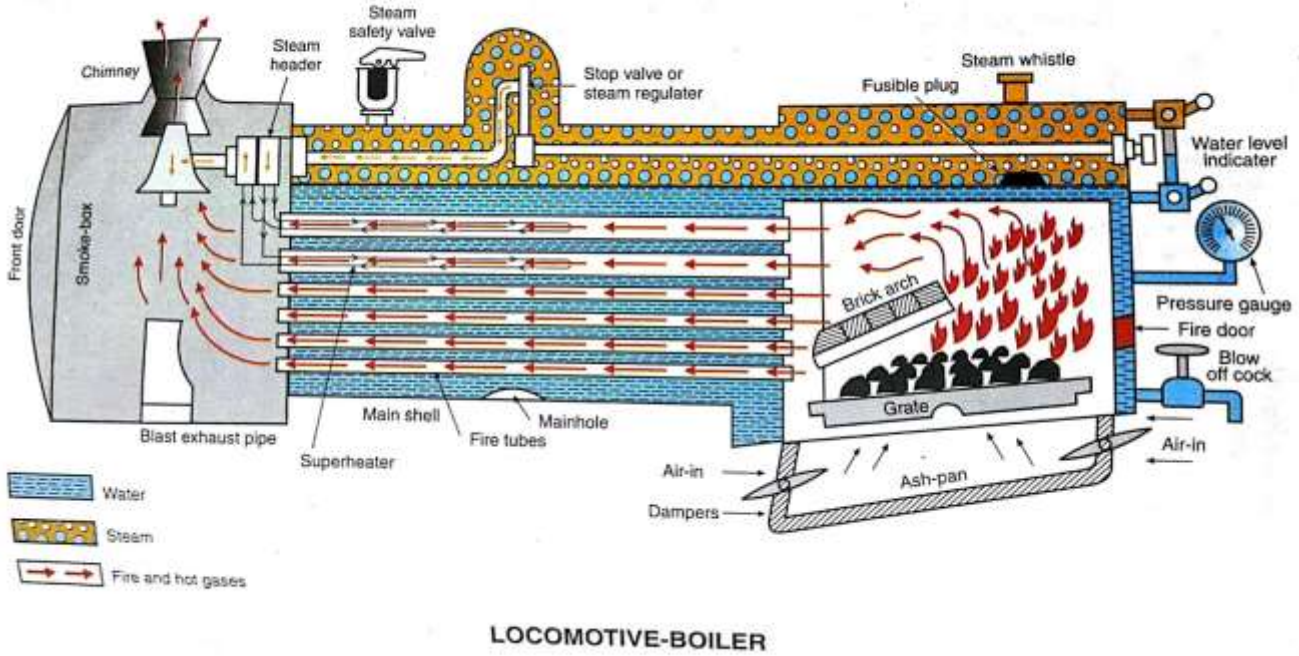
Locomotive Boiler:

This boiler was part of the Locomotive engine in which the steam was used to run the engine vehicle but now this system is powered by fuel which is diesel or electricity. In the early 19's, this locomotive boiler was developed to overcome the railway problem.

The railway engine used a locomotive boiler till the 20th century but now it shifted to fuel and then electricity.

A locomotive Boiler is an external fire boiler. It has a large fire space, horizontal drum axis, and is a multi-tube boiler. It is also a portable type of Boiler.





LOCOMOTIVE-BOILER

Babcock and Wilcox Boiler

The Babcock and Wilcox boiler is invented by George Herman Babcock and Stephen Wilcox in the year 1967. This is a high-pressure and multi-tubular and externally fired water boiler.

The name kept of this boiler because of the inventor's name. This boiler is also producing a high rate of steam almost more than 5000 kg per hour. The overall efficiency of this boiler is good.

The use of Babcock and Wilcox boiler is to produce high steam for use in industry.

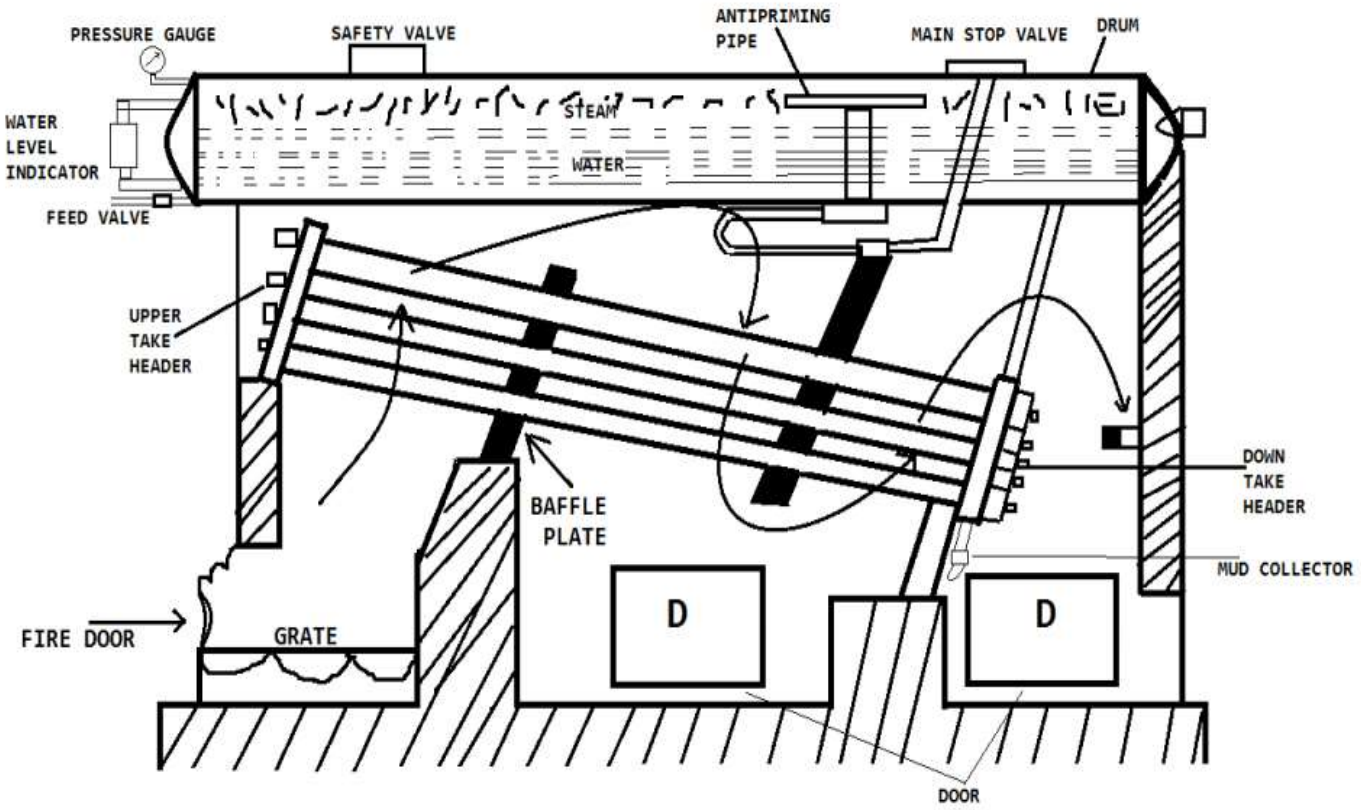


FIG: BABCOCK AND WILCOX BOILER

Water Tube Boiler Advantages:

The following advantages of water tube boiler are:

- The heating surface can be achieved much by using more numbers water tubes.
- The movement of water is much faster than the fire tube boiler because of conventional flow, hence the rate of heat transfer is high which results in higher efficiency.
- Water boiler: Overall efficiency is up to 90%.
- In the water boiler: The maximum working pressure is 250 bar or you can say 140 kg/cm^2.
- The rate of steam generation is high.
- Less floor area required for the given output.
- A Water boiler can be used in a Large Power Plant.
- Load fluctuation or Fluctuation of load can be easily handled.

Water Tube Boiler Disadvantages:

There are some disadvantages of Water boiler:

•The Design is complex in the water boiler.
•For operation, a skilled operator is required.
•The maintenance cost is high.
•The cost of a water tube boiler is high for the same power output.
•As we studied it is used in large power plants therefore it is uneconomical to be used in small industries.
•Feedwater treatment is very essential because small-scale deposits inside the tube cause overheating and bursting.

Water Tube Boiler Application:

The main application of **the water-tube boiler is that it is being used at several places Such as paper mill, Sugar mill, a textile factory, Central heating enterprise, Chemical industry, A power plant, and so on.**

➢ Fire Tube Boiler Advantages:

The following **advantages of fire boiler** are:

•The Fire-tube boiler is in a simple design.
•It is cheaper than the water tube boiler.
•The maintenance cost is low.
•The procedure for cleaning is very simple.
•To operate a fired boiler, no need for skilled operators. Semi-skilled operators can operate.
•A Fire boiler can be used in small industries.
•Feedwater treatment is not very essential.

Fire Tube Boiler Disadvantages:

➢ disadvantages of fire tube boiler:

•The fluctuation of load cannot be easily handled.
•The rate of steam generation is low.
•A Fire boiler is used in small power plants.
•The maximum working pressure is up to 20 bar which is a less comparatively water tube boiler.
•Overall efficiency is up to 75% only.

Fire Tube Boiler Application:

The main application of **Fire Tube Boilers is that it is used in marines**, **railways, and industrial sectors.**

UTILITY BOILERS

- The utility boilers are large capacity steam generators used purely for the electrical power generation.
- An electric utility boiler is defined as a fossil fuel-fired combustion unit of more than 25 megawatts that serves a generator that produces electricity for sale.

Differences between a utility boiler and an industrial boiler

- The size of the boiler
- The application of the steam the boiler generates
- The design of the boiler
- The diversity of the fuels, including the use of by-product fuels
- The global competition for the products created

- A typical utility boiler produces about 3.5 million pounds of steam per hour (approximately 400 MW), while a typical industrial boiler produces about 100,000 pounds of steam per hour.
- Utility boilers are primarily field-erected units designed for high-pressure and high-temperature steam.
- A utility boiler, generates steam for the sole purpose of powering turbines to produce electricity.

